

Readmission Rates:

Predictive Modeling in Diabetic Care

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Introduction

Motivations and Data

Our project predicts whether a patient with diabetes will be readmitted to the hospital within 30 days. Early readmissions signify unresolved discharge issues.

CMS HRRP penalizes hospitals up to 3% of total revenue for excessive early readmissions.

Scale - 101,766 hospital encounters across 130 US hospitals.

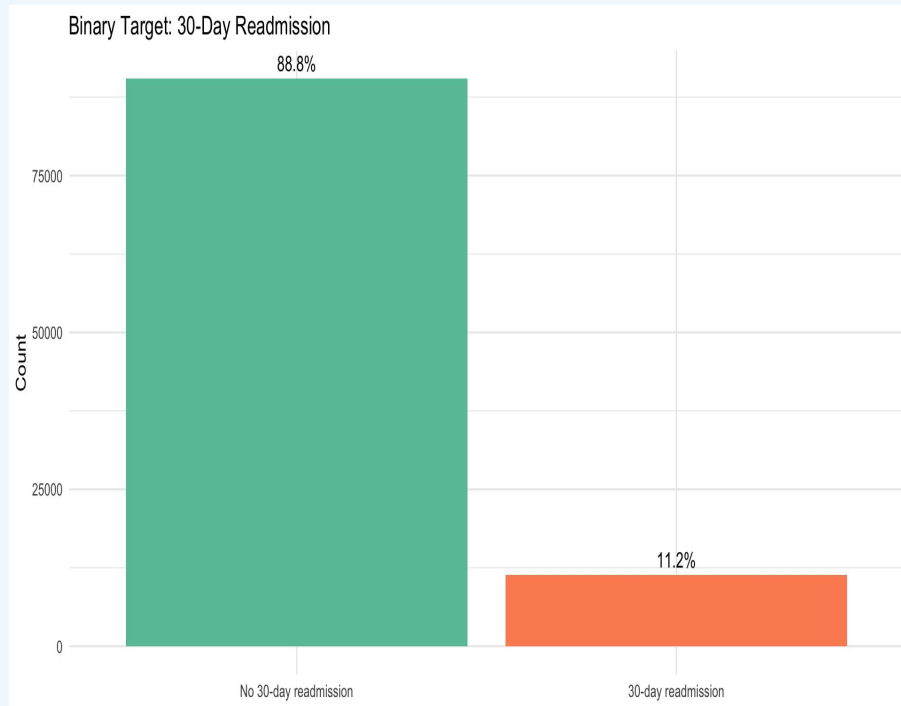
Binary Target & Distribution

The original three-category status (>30, <30, NO) is converted into a binary target

Severe Class Imbalance:

88.8% – No 30-Day Readmission

11.2% – 30 Day Readmission



Why is accuracy is not enough ?

Only 11.2% of patients are 30-day readmissions.

→ Models favors the majority class.

We focus on:

AUC + Sensitivity + Specificity + Balanced Accuracy

The 50 Predictor Variables

Patient Profile

Demographics: Age (10-yr bins), Race, Gender, and Weight.

Encounter: Admission Type/Source, Discharge Destination

Severity and Utilization

Intensity: Length of stay (1-14 days), # of labs, # of procedures.

Utilization: Past 1-year counts of Inpatient, ER, and Outpatient visits

Clinical and Pharmacy Mix

Diagnoses: Primary/ Secondary ICD-9 Codes

Lab Markers: HbA1c results

Medications: 22 diabetes drugs (e.g. Insulin, Metformin) and dosage changes.

Data Preparation/Cleaning

Handling Insignificant Columns and Missing Values

Dropped several unrelated columns:

- Information **unique to each patient** and held no predictive value
- Medications of which over **98%** of patients aren't taking

Feature:	Percent Missing:	Feature Dropped:	Indicators Kept:
Weight	96.9%	TRUE	FALSE
medical_speciality	49.1%	TRUE	TRUE
payer_code	39.6%	TRUE	TRUE

Coding & Final Data Summary

Three diagnosis columns
contain too many ICD-9 codes
-> grouped into fewer
categories to reduce sparsity

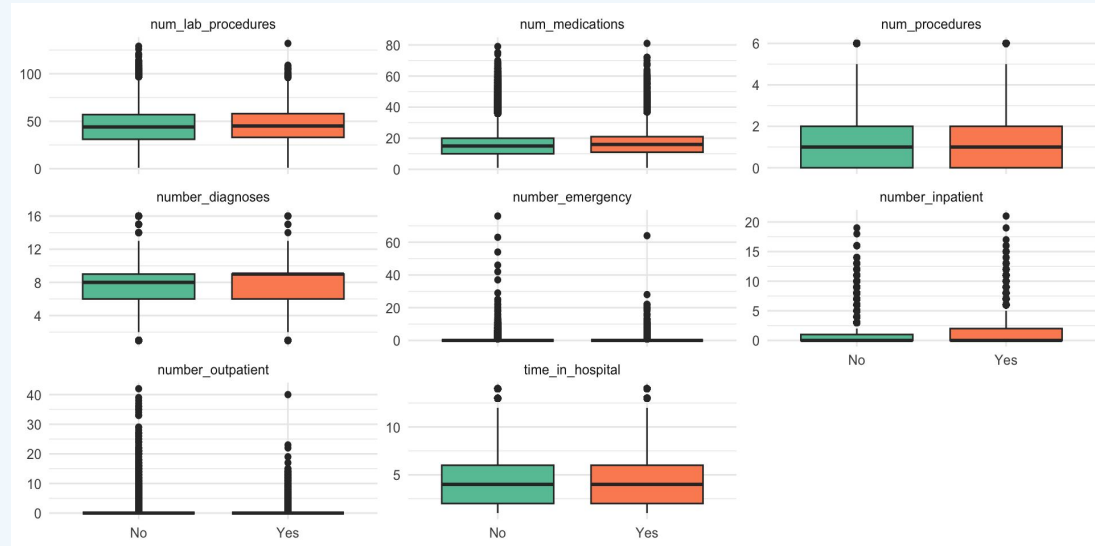
Final dataset dimensions:

- 101,766 rows (patients)
- 32 columns (features - originally 50)

Exploratory Data Analysis

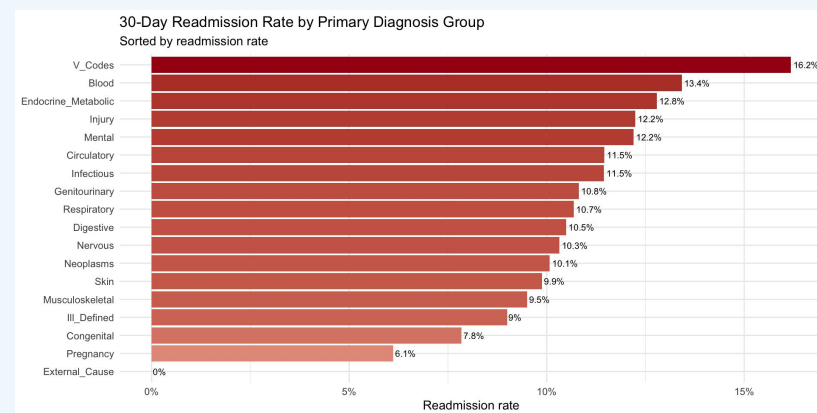
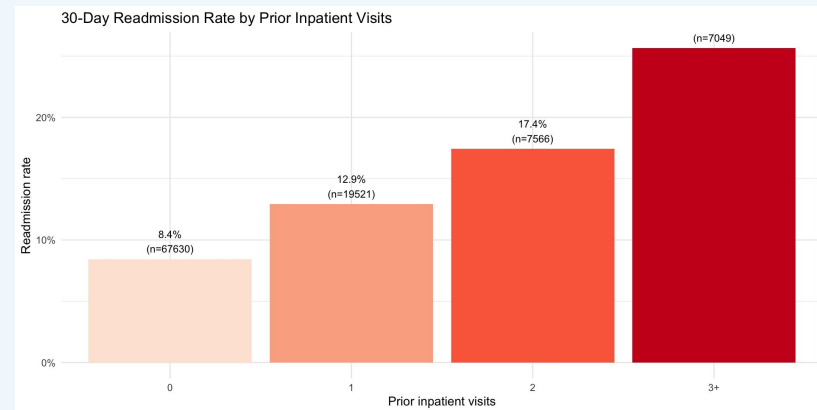
Numerical Predictors

- Predictors split between categorical (race, age group, diagnosis, etc.) and numerical
- `number_inpatient` shows the clearest separation
- Other box plots overlap substantially, showing no strong distinction between groups



Key Risk Factor Profiles

- More prior inpatient visits → higher readmission rate (clear dose-response pattern from 0 to 3+ visits)
- Readmission rates vary by primary diagnosis group (diabetes ranks as the third highest)
- Older age groups generally show higher readmission rates, consistent with greater clinical complexity



Model Selection

Readmission Rates: Model Selection - LASSO Feature selection

154 dummy predictors

(Categorical variables were converted into dummy variables.)



LASSO

(L1 penalty shrinks weak coefficients to zero)

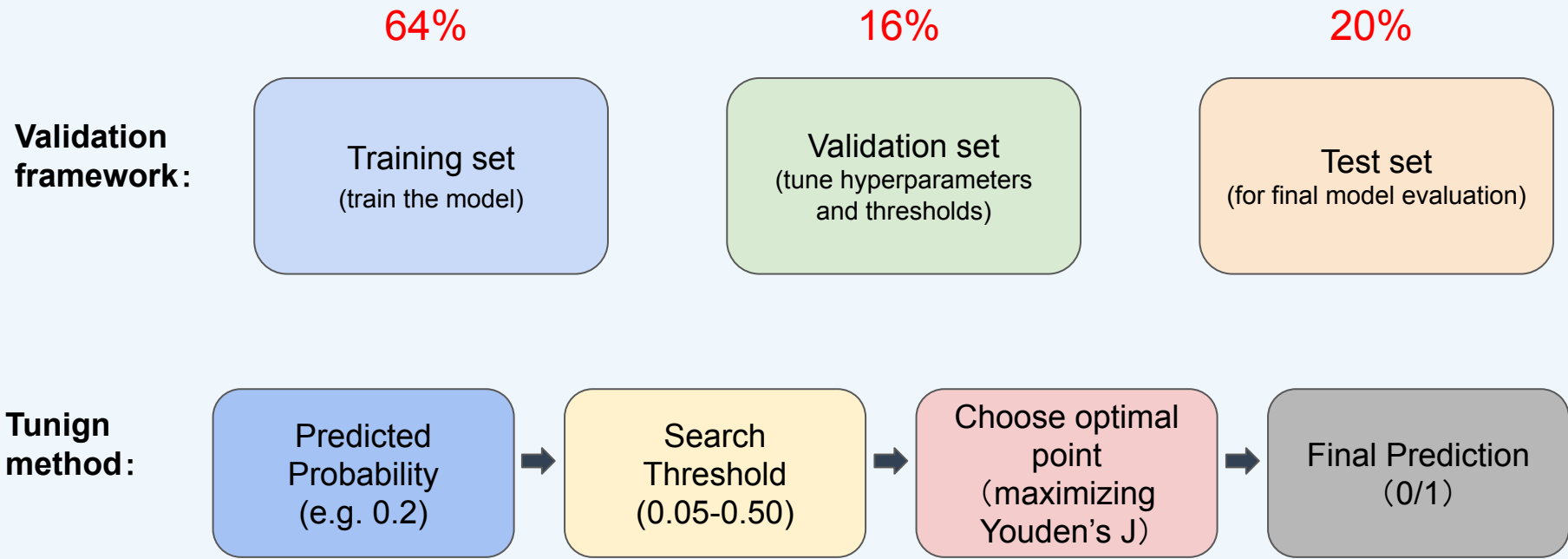
lambda.1se = 0.00422

(a simpler, more stable feature set)



20 dummy predictors

Readmission Rates: Model Selection - Validation & Threshold Tuning



Youden's J = **Specificity** + **Sensitivity** - 1 | Maximizing Balanced Accuracy

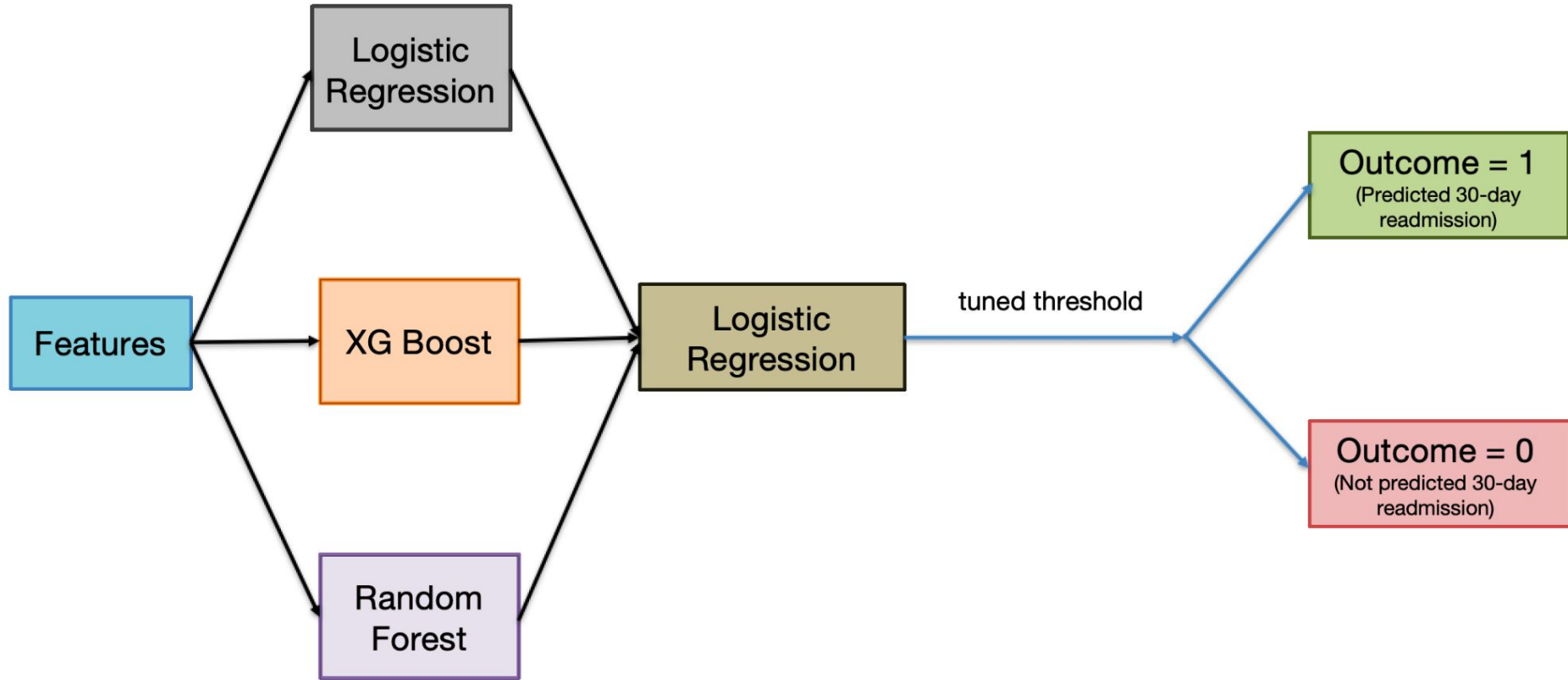
Logistic Regression

Random Forest

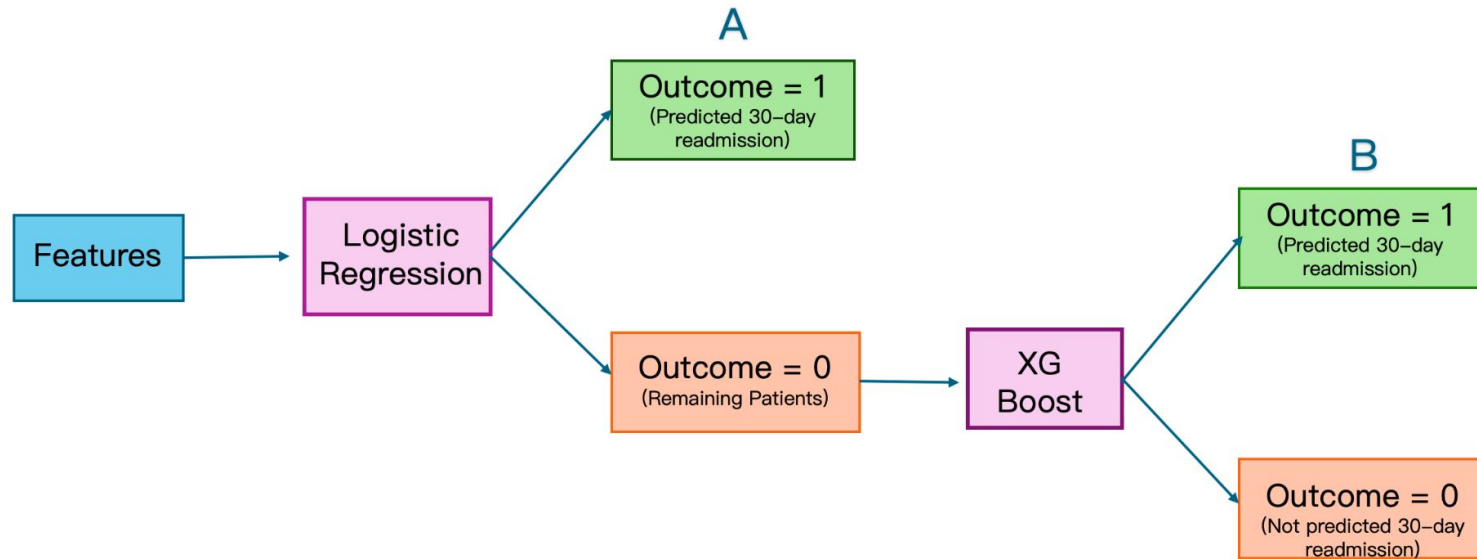
XGBoost

Neural Network

Readmission Rates: Model Selection - Stacking Model



$A+B$ = Final High-Risk Group



Readmission Rates: Model Selection - Model Comparison

Model	AUC	Sensitivity	Specificity	F1	Balanced Acc.
XGBoost	0.673	58.6%	65.3%	27.0%	62.0%
Stacking Ensemble	0.673	61.4%	62.5%	26.7%	62.0%
Two-Stage LR + XGBoost	0.662	91.8%	21.3%	22.4%	56.5%
Logistic Regression	0.662	68.1%	54.9%	25.9%	61.5%
Random Forest	0.656	61.0%	61.1%	25.9%	61.0%
Neural Network	0.649	58.3%	63.7%	26.1%	61.0%

1. **XGBoost** was the best overall model, achieving the **highest** AUC, F1 score, and balanced accuracy.
2. If the priority is to **avoid missing high-risk readmissions**, Two-Stage model can be considered because of its high sensitivity.
3. Logistic Regression remains useful for interpretation(in section 5)

Results

Readmission Risk: Results

Anova table:

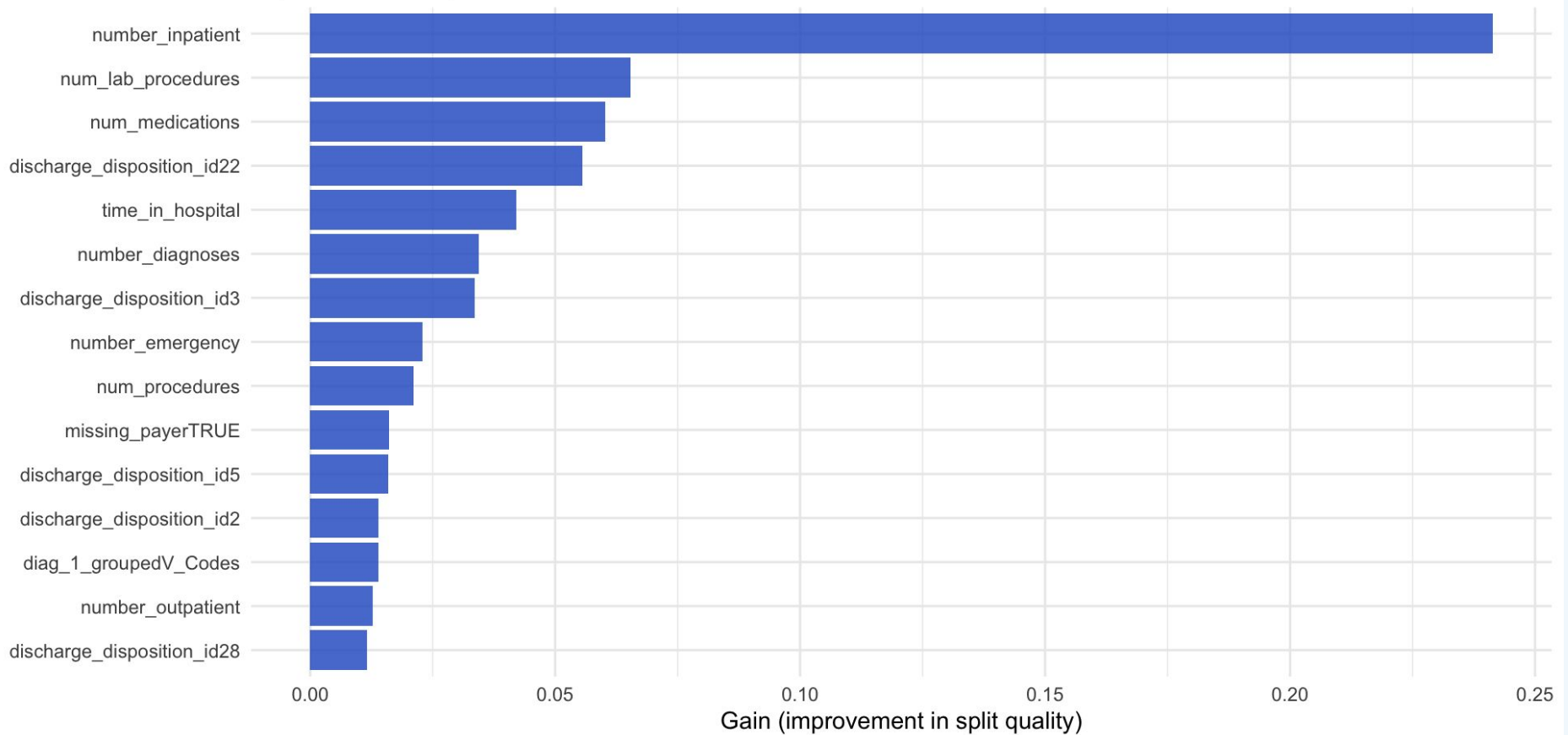
Predictor	LR	Chisq	Df	Pr(>Chisq)
number_inpatient	1.31e+03	1	1.82e-287	
discharge_disposition_id22	4.07e+02	1	1.41e-90	
discharge_disposition_id3	1.67e+02	1	3.70e-38	
discharge_disposition_id5	9.03e+01	1	2.02e-21	
discharge_disposition_id28	7.91e+01	1	5.86e-19	
missing_payerTRUE	4.67e+01	1	8.40e-12	
discharge_disposition_id2	4.57e+01	1	1.41e-11	
number_diagnoses	3.85e+01	1	5.35e-10	
discharge_disposition_id15	3.51e+01	1	3.13e-09	
discharge_disposition_id6	3.49e+01	1	3.42e-09	
diag_2_groupedNeoplasms	2.81e+01	1	1.16e-07	
metforminNo	2.63e+01	1	2.91e-07	
`age[50-60)`	2.57e+01	1	4.05e-07	
diabetesMedYes	2.17e+01	1	3.20e-06	
number_emergency	1.44e+01	1	1.50e-04	
diag_1_groupedIll_Defined	1.12e+01	1	8.01e-04	
diag_3_groupedNeoplasms	9.30e+00	1	2.30e-03	
num_medications	3.36e+00	1	6.67e-02	
insulinNo	2.28e+00	1	1.31e-01	
time_in_hospital	6.54e-03	1	9.36e-01	

Most significant factors:

Factor	Odds Ratio
Prior inpatient hospitalizations	1.31×
Psychiatric hospital discharge	4.93×
Rehabilitation facility discharge	3.78×

Readmission Risk: Results

Top 15 Features by XGBoost Gain



Readmission Risk: Results

SHAP Summary — Impact of Top 15 Features



Conclusion

1. Built an interpretable readmission-risk framework

- Using Relaxed - LASSO to interpret coefficients and odds ratios.
- Using feature importance plot and SHAP plot to identify which variables contributed most to model predictions.

2. Compared multiple predictive models

- Evaluated single-stage models, stacking, and two-stage model under the same validation framework.

1. Improve data preprocessing

- Further refine missing-value handling, diagnosis grouping

2. Conduct stronger model robustness checks

- Test alternative feature-selection settings, threshold rules, and hyperparameter ranges to see whether model rankings remain stable.

3. Validate on an external dataset

- Evaluate the final model on data from a different hospital system to test generalizability.

We used AI tools, including Claude Code, Codex, and ChatGPT, to assist with code generation, debugging, graph creation, and doing repetitive model tuning procedures.

THANK YOU!